Discovery

- GSE165816 scRNA-seq: 54 libraries from 27 people; 44 skin libraries, 131,507 cells
- Uniform QC (200-5,000 genes, <10% mito): 97,677 skin cells, no sub-sampling
- Harmony integration, Leiden clustering, post hoc annotation; myeloid re-clustering (7,756 cells)
- Primary contrast: macrophage-state composition and program scores, 12 healing vs 7 non-healing libraries
- Library = unit of analysis; raw-count pseudobulk; model ~ sex + tissue site + group (Figs 2-6)

Internal sensitivity analyses

- Cell-level vs library-level testing; QC-threshold grid; trajectory rooted at either end
- Covariate models (sex, site, depth, %mito); foot-only (14 libraries); putative-donor aggregation (12 donors)
- Exclusion of libraries with mixed XIST/Y signal; removal of opposite-sex cells
- Classifier: library-grouped, leave-one-library-out and donor-grouped CV; permutation nulls; bootstrap CIs
- Deconvolution benchmark on 200 synthetic mixtures; power calculation (Figs 6, 10, 13, 15)

External transportability testing

- GSE134431 foot ulcers (13; bulk): corroboration only - used in panel construction, not independent
- GSE143735 forearm skin (9; bulk): untouched external test of direction, effect size and classifier transfer
- Directional concordance vs statistical replication (same sign and $P < 0.05$), with exact 95% CIs (Figs 9, 12, 14)

Hypothesis-generating in-silico analyses

- Ligand-receptor communication; trajectory gene waves; co-expression networks (Figs 7, 8, 11)
- 129-gene candidate panel = genes recurring in >= 3 of 6 discovery evidence sets (Supplementary Table 5)
- In-silico perturbation of the fitted classifier: associative model responses, no causal meaning (Figs 16, 17)
- Not inferential evidence; candidates for foot-matched multiplex IF, spatial, flow and functional assays

Interpretation: exploratory; directionally consistent findings are not treated as established patient-level differences